

SECURE INFRASTRUCTURE DEPLOYMENT AND MONITORING

Final Practical Assignment

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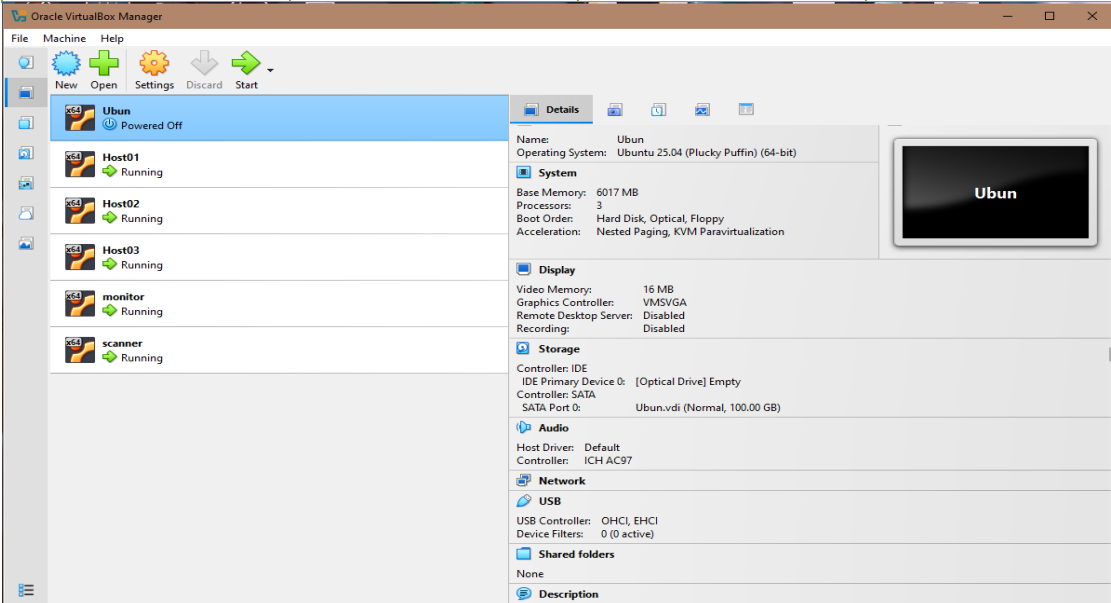
Introduction

This report documents the deployment of a secure virtual infrastructure using Oracle VirtualBox, Ubuntu Server, Wazuh and OpenVAS. The goal was to build a monitored enterprise network suitable for vulnerability assessment and centralized security monitoring.

1.1 Virtual Infrastructure

Five Ubuntu virtual machines were deployed: one monitoring server (Wazuh), one vulnerability scanner (OpenVAS) and three monitored hosts.

VM	Role	OS	IP
Monitor	Wazuh	Ubuntu	192.168.32.210
Scanner	OpenVAS	Ubuntu	192.168.32.220
Host01	Client	Ubuntu	192.168.32.201
Host02	Client	Ubuntu	192.168.32.202
Host03	Client	Ubuntu	192.168.32.203



1.2 Network Configuration

Static IP addresses were assigned to every virtual machine. Connectivity was verified with ping commands between all hosts.

- ping 192.168.32.201
- ping 192.168.32.202
- ping 192.168.32.203
- ping 192.168.32.210
- ping 192.168.32.220

```
host@monitor: $ ping google.com
PING google.com (2a00:1450:4025:807::8b) 56 data bytes
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=1 ttl=114 time=39.0 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=2 ttl=114 time=54.1 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=3 ttl=114 time=48.5 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=4 ttl=114 time=34.4 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=5 ttl=114 time=38.8 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=6 ttl=114 time=113 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=7 ttl=114 time=41.3 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=8 ttl=114 time=53.8 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=9 ttl=114 time=25.5 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=10 ttl=114 time=29.3 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=11 ttl=114 time=56.0 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=12 ttl=114 time=61.1 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=13 ttl=114 time=65.8 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=14 ttl=114 time=52.3 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=15 ttl=114 time=45.2 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=16 ttl=114 time=44.9 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=17 ttl=114 time=44.9 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=18 ttl=114 time=33.0 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=19 ttl=114 time=34.9 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=20 ttl=114 time=38.1 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=21 ttl=114 time=24.3 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=22 ttl=114 time=71.4 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=23 ttl=114 time=34.8 ms
64 bytes from nt-in-f139.1e100.net (2a00:1450:4025:807::8b): icmp_seq=24 ttl=114 time=37.0 ms
^C
--- google.com ping statistics ---
24 packets transmitted, 24 received, 0% packet loss, time 23031ms
rtt min/avg/max/mdev = 24.336/46.738/113.374/18.314 ms
```

1.3 Host Configuration

Each VM received a unique hostname and the OpenSSH server was enabled for secure remote administration.

```
sudo hostnamectl set-hostname host01
```

```
sudo systemctl enable ssh
```

```
sudo systemctl start ssh
```

```
PS C:\Users\Kalyte> ssh host@192.168.32.201
host@192.168.32.201's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Jul 19 11:32:33 AM UTC 2026

System load:            0.0
Usage of /:             28.7% of 23.39GB
Memory usage:          11%
Swap usage:            0%
Processes:             123
Users logged in:       0
IPv4 address for enp0s3: 192.168.32.201
IPv6 address for enp0s3: 2a00:1eb8:c0d3:f22c:a00:27ff:fe3f:5149

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Jul 19 11:21:04 2026 from 192.168.32.179
```

Summary

The virtual infrastructure was successfully deployed and prepared for the following stages: infrastructure hardening, Wazuh monitoring and OpenVAS vulnerability scanning.

Task 2 - INFRASTRUCTURE HARDENING

This section describes the hardening techniques applied to the Ubuntu infrastructure in order to improve security before deploying monitoring services.

2.1 System Updates

All Ubuntu virtual machines were updated before additional software was installed. Updating the operating system ensures that known vulnerabilities are patched.

```
sudo apt update && sudo apt upgrade -y
```

```
host@monitor:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://lt.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://lt.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://lt.archive.ubuntu.com/ubuntu resolute-backports InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
host@monitor:~$ sudo apt upgrade -y
Not upgrading yet due to phasing:
python3-software-properties software-properties-common

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
host@monitor:~$
```

2.2 SSH Hardening

Secure Shell was configured for remote administration. Administrative access should be restricted to authorized users only. Password authentication may be replaced with SSH keys in production environments.

Setting	Recommendation
PermitRootLogin	no
PasswordAuthentication	yes / SSH keys preferred
Port	22 (or custom in production)
AllowUsers	Administrator accounts only

```
host@monitor:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-07-19 15:52:25 UTC; 4h 32min ago
   Invocation: e6850408b2934ac99dda7e987f26d537
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 1311 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1340 (sshd)
     Tasks: 1 (limit: 5145)
    Memory: 8M (peak: 9.6M)
       CPU: 130ms
   CGroup: /system.slice/ssh.service
           └─1340 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 19 15:52:24 monitor systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jul 19 15:52:25 monitor sshd[1340]: Server listening on 0.0.0.0 port 22.
Jul 19 15:52:25 monitor sshd[1340]: Server listening on :: port 22.
Jul 19 15:52:25 monitor systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jul 19 15:52:47 monitor sshd-session[2932]: Accepted password for host from 192.168.32.202 port 47074 ssh2
Jul 19 15:52:47 monitor sshd-session[2932]: pam_unix(sshd:session): session opened for user host(uid=1000) by host(uid=0)
Jul 19 16:08:41 monitor sshd-session[10362]: Accepted publickey for host from 192.168.32.201 port 54738 ssh2: ED25519 SHA256:KOCsY7n0wISgwfd5FZUTQgDg9%
Jul 19 16:08:41 monitor sshd-session[10362]: pam_unix(sshd:session): session opened for user host(uid=1000) by host(uid=0)
Jul 19 16:30:26 monitor sshd-session[10991]: Accepted publickey for host from 192.168.32.201 port 38516 ssh2: ED25519 SHA256:KOCsY7n0wISgwfd5FZUTQgDg9%
Jul 19 16:30:26 monitor sshd-session[10991]: pam_unix(sshd:session): session opened for user host(uid=1000) by host(uid=0)
log file: s -teln | grep :22_
```

2.3 Firewall Configuration

The Uncomplicated Firewall (UFW) was enabled to restrict unnecessary inbound connections.

- `sudo ufw allow OpenSSH`
- `sudo ufw allow 1514/tcp`
- `sudo ufw allow 1515/tcp`
- `sudo ufw enable`
- `sudo ufw status`

```
host@host01:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
host@host01:~$ sudo ufw status
Status: active
host@host01:~$
```

2.4 Password Policy

Strong passwords were used for administrative accounts. Passwords should contain uppercase and lowercase letters, numbers and special characters. Default credentials should always be changed after installation.

Task 2 Summary

The infrastructure hardening phase reduced the attack surface of the virtual environment by applying software updates, securing remote access, enabling firewall protection and introducing centralized configuration management concepts.

```
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-28-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Mon Jul 20 04:25:13 PM UTC 2026

System load:            0.14
Usage of /:             28.8% of 23.39GB
Memory usage:          10%
Swap usage:            0%
Processes:             120
Users logged in:       0
IPv4 address for enp0s3: 192.168.32.202
IPv6 address for enp0s3: 2a00:1eb8:c0d3:f22c:a00:27ff:fe58:ac29

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Jul 19 06:14:52 2026 from 192.168.32.201
host@host02: ~$ sudo systemctl status ssh
[sudo: authenticate] Password:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-07-20 15:40:39 UTC; 44min ago
   Invocation: dedf3184d7664301a67d885544c06238
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 1267 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1312 (sshd)
     Tasks: 1 (limit: 1718)
    Memory: 5.7M (peak: 7.2M)
       CPU: 76ms
    CGroup: /system.slice/ssh.service
            └─1312 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

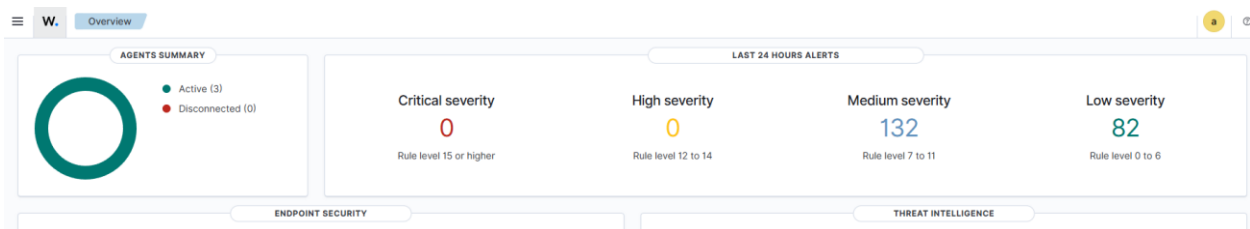
Jul 20 15:40:38 host02 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jul 20 15:40:39 host02 sshd[1312]: Server listening on 0.0.0.0 port 22.
Jul 20 15:40:39 host02 sshd[1312]: Server listening on :: port 22.
Jul 20 15:40:39 host02 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jul 20 16:25:13 host02 sshd-session[2329]: Accepted password for host from 192.168.32.179 port 59881 ssh2
Jul 20 16:25:13 host02 sshd-session[2329]: pam_unix(sshd:session): session opened for user host(uid=1000) by host(uid=0)
host@host02: ~$
```

Task 3 - WAZUH DEPLOYMENT AND SECURITY MONITORING

This section describes the deployment of the Wazuh platform and the integration of monitored endpoints. Wazuh provides centralized log collection, security monitoring, file integrity monitoring and threat detection.

3.1 Wazuh Server Deployment

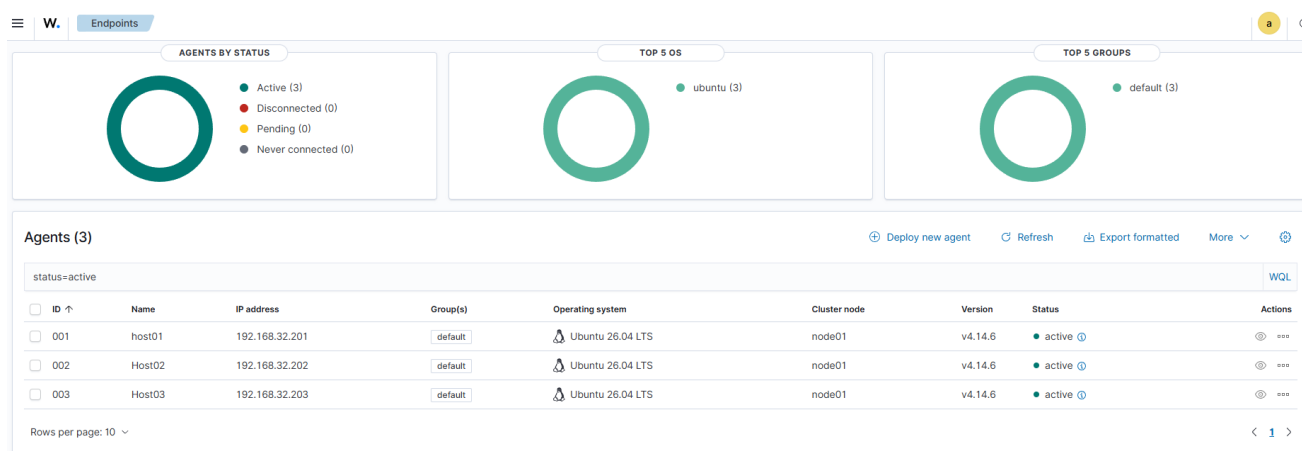
The Wazuh server was installed on the dedicated monitoring virtual machine. The deployment included the Wazuh Manager, Indexer and Dashboard components. After installation, all services were verified to ensure correct operation.



3.2 Wazuh Agent Installation

Wazuh agents were installed on all monitored Ubuntu hosts. Each endpoint was registered with the Wazuh Manager and appeared as an active agent in the dashboard.

Hostname	IP Address	Status
host01	192.168.32.201	Active
host02	192.168.32.202	Active
host03	192.168.32.203	Active



3.3 Event Monitoring

The dashboard was used to monitor authentication events, system activity and security alerts in real time. Events generated by the monitored hosts were collected, indexed and displayed through the centralized interface.

Figure 3.3 – Wazuh Dashboard Showing Active Agents

3.6 Security Benefits

Feature	Benefit
Centralized Monitoring	Single management interface
Log Collection	Simplified incident investigation
File Integrity Monitoring	Detects unauthorized changes
Security Alerts	Early threat detection
Agent Management	Centralized endpoint administration

Task 3 Summary

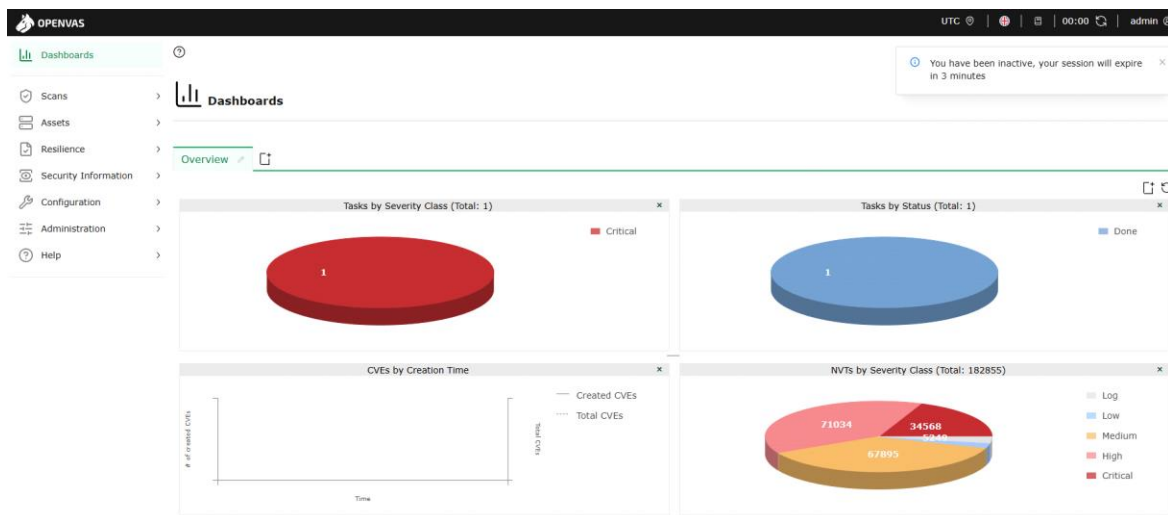
The Wazuh deployment successfully provided centralized visibility across the laboratory environment. All monitored hosts reported security events to the manager, allowing administrators to detect suspicious activities, review logs and monitor endpoint status from a single dashboard.

Task 4 - OPENVAS VULNERABILITY ASSESSMENT

This section presents the vulnerability assessment performed using Greenbone OpenVAS together with the final evaluation of the project.

4.1 OpenVAS Deployment

Greenbone Community Edition (OpenVAS) was deployed on a dedicated Ubuntu virtual machine using Docker containers. After installation, the web interface was used to configure scan targets and execute vulnerability assessments.



4.2 Target Configuration

The monitored hosts were added as scan targets using their static IP addresses. Each target was configured individually to ensure accurate scan results.

Target	IP Address	Purpose
host01	192.168.32.201	Endpoint
host02	192.168.32.202	Endpoint
host03	192.168.32.203	Endpoint

OPENVAS

UTC 00:00 admin

Dashboards

Scans

Assets

Resilience

Security Information

Configuration

Targets

Port Lists

Credentials

Scan Configs

Alerts

Schedules

Report Configs

Report Formats

Scanners

Filters

Tags

Administration

Help

Feed is currently syncing.

Please wait while the feed is syncing. Scans are not available during this time. For more information, visit the [Documentation](#).

You have been inactive, your session will expire in 3 minutes

Filter

Targets 2 of 2

Name	Hosts	IPs	Port List	Credentials	Actions
Local Network	192.168.32.0/24	254	All IANA assigned TCP		
Unnamed	192.168.32.0/24	254	All IANA assigned TCP		

Apply to page contents

(Applied filter: sort=name first=1 count=10)

4.3 Scan Results

The vulnerability scans identified installed services, open ports and security weaknesses. The generated reports allow administrators to prioritize remediation activities according to risk severity.

OPENVAS

UTC 13:17 admin

Dashboards

Scans

Tasks

Reports

Results

Vulnerabilities

Notes

Overrides

Assets

Resilience

Security Information

Configuration

Administration

Help

Vulnerabilities 44 of 47

Vulnerabilities by CVSS (Total: 44)

Vulnerabilities by Severity Class (Total: 44)

Name	Oldest Result	Newest Result	Severity	QoD	Results	Hosts
MySQL / MariaDB Default Credentials (MySQL Protocol)	Sat, Jul 18, 2026 10:10 PM Coordinated Universal Time	Sun, Jul 19, 2026 12:35 AM Coordinated Universal Time	9.8 (Critical)	95 %	2	1
Embedthis GoAhead 2.5.0 HTTP Header Injection Vulnerability - Active Check	Sat, Jul 18, 2026 11:22 PM Coordinated Universal Time	Sun, Jul 19, 2026 1:04 AM Coordinated Universal Time	8.6 (High)	99 %	4	1
DCE/RPC and MSRPC Services Enumeration Reporting	Sat, Jul 18, 2026 10:09 PM Coordinated Universal Time	Sun, Jul 19, 2026 12:33 AM Coordinated Universal Time	5.0 (Medium)	80 %	2	1
TCP Timestamps Information Disclosure	Sat, Jul 18, 2026 10:00 PM Coordinated Universal Time	Sun, Jul 19, 2026 12:31 AM Coordinated Universal Time	2.6 (Low)	80 %	10	5
Weak MAC Algorithm(s) Supported (SSH)	Sat, Jul 18, 2026 10:03 PM Coordinated Universal Time	Sun, Jul 19, 2026 12:35 AM Coordinated Universal Time	2.6 (Low)	80 %	10	5

4.4 Risk Analysis

Severity	Description	Recommended Action
Critical	Immediate exploitation possible	Patch immediately
High	High security risk	Update affected software
Medium	Potential weakness	Review and mitigate
Low	Minor issue	Monitor and document
Info	Informational finding	No immediate action

4.5 Remediation

The recommended remediation actions include applying operating system updates, removing unnecessary services, strengthening authentication mechanisms, enabling firewall protection and continuously monitoring the environment using Wazuh.

Conclusion

The objectives of the practical assignment were successfully achieved. A secure virtual infrastructure was deployed, hardened, monitored using Wazuh and assessed with OpenVAS. The implemented solution demonstrates the integration of system administration, security monitoring and vulnerability management in a virtual enterprise environment.

References

- Oracle VM VirtualBox Documentation
- Ubuntu Server Documentation
- Wazuh Documentation
- Greenbone Community Edition Documentation
- OpenSSH Documentation
- UFW Documentation